



JABchem



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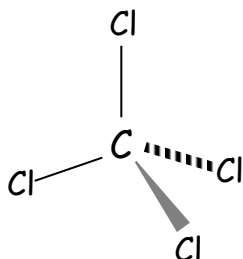
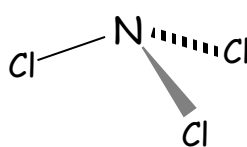
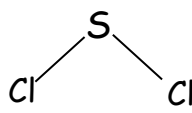
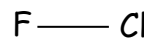
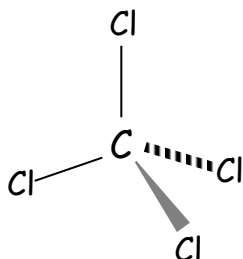
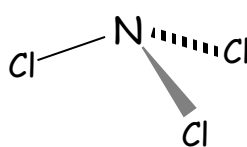
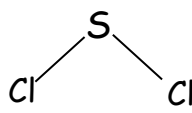
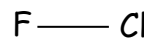
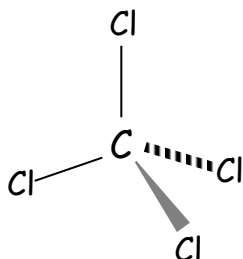
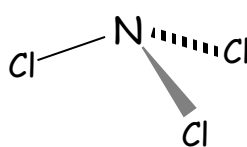
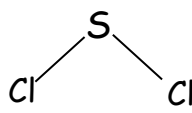
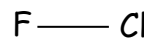
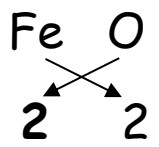
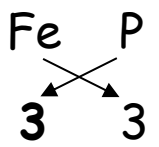
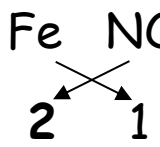
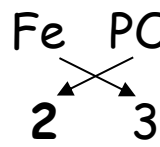
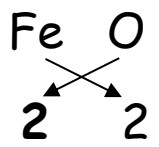
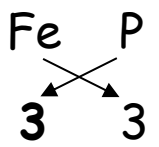
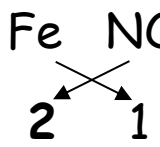
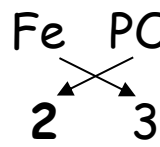
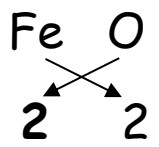
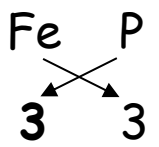
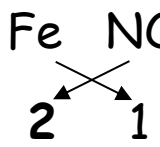
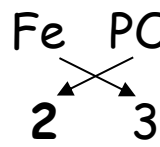
Past Papers Nat 5 Chemistry

2023 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/100)	%	
A	69+	69%	42.5%
B	57+	57%	20.3%
C	46+	46%	15.1%
D	34+	34%	12.2%
No award	<34	<34%	9.9%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.1 /25	46.0 /75	No Assignment in 2023

2023 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning																
1	D	62	$\text{Rate} = \frac{\Delta \text{Quantity}}{\Delta \text{Time}} = \frac{300 \text{ cm}^3}{60 \text{ s}} = 5 \text{ cm}^3 \text{ s}^{-1}$																
2	A	92	<table border="1"> <thead> <tr> <th>Particle</th><th>Location</th><th>Charge</th><th>Mass</th></tr> </thead> <tbody> <tr> <td>proton</td><td>inside nucleus</td><td>+1</td><td>1 amu</td></tr> <tr> <td>neutron</td><td>inside nucleus</td><td>0</td><td>1 amu</td></tr> <tr> <td>electron</td><td>outside nucleus</td><td>-1</td><td>approx 0</td></tr> </tbody> </table>	Particle	Location	Charge	Mass	proton	inside nucleus	+1	1 amu	neutron	inside nucleus	0	1 amu	electron	outside nucleus	-1	approx 0
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3	C	80	<table border="1"> <thead> <tr> <th>A</th><th>B</th><th>C</th><th>D</th></tr> </thead> <tbody> <tr> <td> CCl_4  </td><td> NCl_3  </td><td> SCl_2  </td><td> FCl  </td></tr> <tr> <td>tetrahedral</td><td>trigonal pyramidal</td><td>angular</td><td>linear</td></tr> </tbody> </table>	A	B	C	D	CCl_4 	NCl_3 	SCl_2 	FCl 	tetrahedral	trigonal pyramidal	angular	linear				
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CCl_4 	NCl_3 	SCl_2 	FCl 																
tetrahedral	trigonal pyramidal	angular	linear																
4	B	58	Lithium Chloride is ionic as compound has metal bonded to non-metal. ☒ A Diagram shows metallic bonding (positive ions surrounded by delocalised electrons) ☒ B Diagram shows ionic bonding (positive ions and negative ions) ☒ C Diagram shows covalent molecular (separate molecules with covalent bonds inside molecule) ☒ D Diagram shows covalent network (long network of covalent bonds)																
5	B	52	<table border="1"> <thead> <tr> <th>A</th><th>B</th><th>C</th><th>D</th></tr> </thead> <tbody> <tr> <td> Fe O  </td><td> Fe P  </td><td> Fe NO_3^-  </td><td> Fe PO_4^{3-}  </td></tr> <tr> <td>valency of Fe = 2</td><td>valency of Fe = 3</td><td>valency of Fe = 2</td><td>valency of Fe = 2</td></tr> </tbody> </table>	A	B	C	D	Fe O 	Fe P 	Fe NO_3^- 	Fe PO_4^{3-} 	valency of Fe = 2	valency of Fe = 3	valency of Fe = 2	valency of Fe = 2				
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valency of Fe = 2	valency of Fe = 3	valency of Fe = 2	valency of Fe = 2																
6	D	71	☒ A no. of mol = volume x concentration = $0.1 \text{ litres} \times 1.00 \text{ mol l}^{-1} = 0.1 \text{ mol}$ ☒ B no. of mol = volume x concentration = $0.15 \text{ litres} \times 0.75 \text{ mol l}^{-1} = 0.1125 \text{ mol}$ ☒ C no. of mol = volume x concentration = $0.2 \text{ litres} \times 0.60 \text{ mol l}^{-1} = 0.12 \text{ mol}$ ☒ D no. of mol = volume x concentration = $0.25 \text{ litres} \times 0.25 \text{ mol l}^{-1} = 0.0625 \text{ mol}$																
7	B	54	☒ A aluminium oxide is insoluble $\therefore$ no change to pH of water ☒ B barium oxide is a soluble metal oxide $\therefore$ dissolves in water to form alkaline pH>7 ☒ C nitrogen oxide is a non-metal oxide $\therefore$ dissolves in water to form pH<7 ☒ D hydrogen oxide H_2O is water with a pH=7 $\therefore$ does not change pH of water when added																
8	C	45	nickel carbonate + sulfuric acid $\longrightarrow$ nickel sulfate + water + carbon dioxide nickel hydroxide + sulfuric acid $\longrightarrow$ nickel sulfate + water nickel + sulfuric acid $\longrightarrow$ nickel sulfate + hydrogen ☒ A nickel hydroxide does not produce a gas when reacted with sulfuric acid ☒ B nickel does not produce a gas when reacted with sulfuric acid ☒ C nickel sulfate is produced in all three reactions ☒ D no water produced in reaction of nickel and sulfuric acid $\therefore$ not a neutralisation reaction																

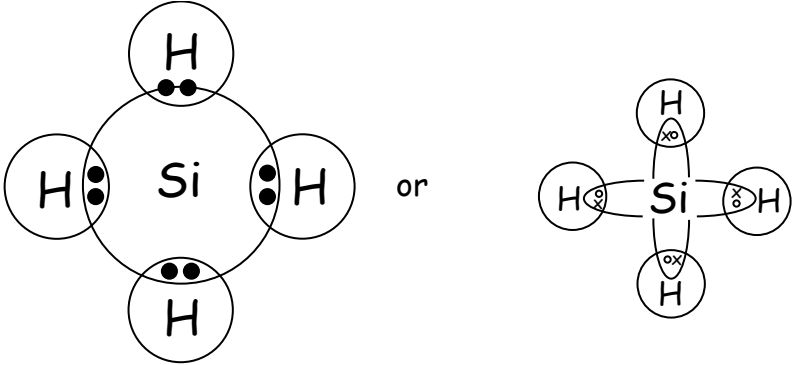
9	B	59	$(\text{Na}^+)_2\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+\text{Cl}^-(\text{aq}) \rightarrow 2\text{Na}^+\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ Split solutions into ions $2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ Identify Spectator Ions and cancel out Spectator Ions $2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ Re-write equation without spectator ions $\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$															
10	A	49	Carbon number: 5 4 3 2 1 $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{C}(\text{CH}_3)_2\text{CH}_3$ ↑ 1x methyl group on C₄ ↑ 2x methyl group on C₂ Name of compound: 2,2,4-trimethylpentane NB: Overall formula is of alkane as formula C₈H₁₈ fits general formula C_nH_{2n+2}															
11	C	56	☒ A This structure is also 2-methylbut-2-ene but drawn differently ☒ B This structure has formula C ₅ H ₁₂ but 2-methylbut-2-ene has formula C ₅ H ₁₀ ☒ C Both have formula C ₅ H ₁₀ and have different structures ∴ isomers ☒ D This structure has formula C ₆ H ₁₂ but 2-methylbut-2-ene has formula C ₅ H ₁₀															
12	B	33	☒ A Addition of H ₂ across the C=C double bond in but-2-ene produces butane C ₄ H ₁₀ ☒ B Butan-1-ol cannot be produced as -OH group must be added to C ₂ to form butan-2-ol ☒ C Addition of H ₂ O across the C=C double bond in but-2-ene produces butan-2-ol ☒ D Addition of Br ₂ across the C=C double bond in but-2-ene produces 2,3-dibromobutane															
13	A	87																
14	D	68	☒ A methanol CH ₃ OH has a lower formula mass compared to octan-1-ol C ₈ H ₁₇ OH ☒ B CH ₃ OH has a higher solubility than C ₈ H ₁₇ OH as methanol has a shorter carbon chain ☒ C methanol CH ₃ OH has a lower formula mass compared to octan-1-ol C ₈ H ₁₇ OH ☒ D methanol CH ₃ OH has a lower formula mass and higher solubility than octan-1-ol C ₈ H ₁₇ OH															
15	D	78	<table><tr><th>Chemistry</th><th>butane</th><th>cyclobutane</th></tr><tr><td>General Formula</td><td>C₄H₁₀ fits C_nH_{2n+2}</td><td>C₄H₈ fits C_nH_{2n}</td></tr><tr><td>Melting Point</td><td>-138°C</td><td>-90°C</td></tr><tr><td>Solubility in water</td><td>insoluble in water</td><td>insoluble in water</td></tr><tr><td>Saturation</td><td>Saturated (no C=C double bonds)</td><td>Saturated (no C=C double bonds)</td></tr></table>	Chemistry	butane	cyclobutane	General Formula	C ₄ H ₁₀ fits C _n H _{2n+2}	C ₄ H ₈ fits C _n H _{2n}	Melting Point	-138°C	-90°C	Solubility in water	insoluble in water	insoluble in water	Saturation	Saturated (no C=C double bonds)	Saturated (no C=C double bonds)
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16	C	69	Acid + Metal Oxide → Salt + Water methanoic acid + sodium oxide → sodium methanoate + water															
17	A	72	☒ A conducts as solid and liquid ∴ metallic bonding ☒ B does not conduct as solid or liquid and has low m.pt. and b.pt. ∴ covalent molecular ☒ C conducts as liquid but not solid ∴ ionic bonding ☒ D does not conduct as solid or liquid and has high m.pt. ∴ covalent network															
18	B	70	☒ A Y is least reactive as it is only metal not to react with acid. (Least reactive comes first in list) ☒ B Y is least reactive and Z is most reactive ∴ Y then X then Z ☒ C Z is most reactive as it is only metal to react with water. (Most reactive comes last in list) ☒ D Z is most reactive as it is only metal to react with water. (Most reactive comes last in list)															

19	D	86	☒ A Nickel(A) is lower than zinc(B) in Electrochemical series so electrons flow from B to A ☒ B Zinc(A) is lower than aluminium(B) in Electrochemical series so electrons flow from B to A ☒ C Aluminium(A) is lower than magnesium(B) in Electrochemical series so electrons flow from B to A ☒ D Aluminium(A) is higher than nickel(B) in Electrochemical series so electrons flow from A to B			
20	A	29	As Magnesium is higher up the electrochemical series. Magnesium will oxidise into Mg^{2+} ions and Ag^+ ions will reduce back to silver atoms in a redox reaction. Oxidation $Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$ Reduction $Ag^+(aq) + e^- \longrightarrow Ag(s)$ Oxidation $Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$ Reduction x2 $2Ag^+(aq) + 2e^- \longrightarrow 2Ag(s)$ Redox $Mg(s) + 2Ag^+(aq) \longrightarrow Mg^{2+}(aq) + 2Ag(s)$			
21	C	72	Process	Reactants	Products	Catalyst
			Haber	Nitrogen + Hydrogen	Ammonia	Iron
			Ostwald	Ammonia + oxygen	Nitric acid (ultimately)	Platinum
22	C	74	${}_{90}^{227}Th \xrightarrow{\alpha} {}_{88}^{223}Ra \xrightarrow{\alpha} {}_{86}^{219}Rn \xrightarrow{\alpha} {}_{84}^{215}Po \xrightarrow{\alpha} {}_{82}^{211}Pb$			
23	C	70	☒ A Barium sulfate is insoluble and can be collected in a precipitation reaction by filtration ☒ B Lead(II) sulfate is insoluble and can be collected in a precipitation reaction by filtration ☒ C Calcium chloride is soluble so will not form a precipitate ☒ D Silver Chloride is insoluble and can be collected in a precipitation reaction by filtration			
24	A	80	☒ A Glucose gives blue to orange with Benedict's and sodium chloride gives yellow flame test ☒ B Starch is not present as there was no change in the iodine test ☒ C Flame test gave yellow flame but strontium chloride would give a red flame test result ☒ D Starch is not present as there was no change in the iodine test			
25	B	44	$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$ 2mol 0.004mol 1mol 0.002mol			

2023 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning																
1a	7	<table><tr><td>Diatomic Element</td><td>Hydrogen</td><td>Nitrogen</td><td>Oxygen</td><td>Fluorine</td><td>Chlorine</td><td>Bromine</td><td>Iodine</td></tr><tr><td>Formula</td><td>H₂</td><td>N₂</td><td>O₂</td><td>F₂</td><td>Cl₂</td><td>Br₂</td><td>I₂</td></tr></table>	Diatomic Element	Hydrogen	Nitrogen	Oxygen	Fluorine	Chlorine	Bromine	Iodine	Formula	H ₂	N ₂	O ₂	F ₂	Cl ₂	Br ₂	I ₂
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Formula	H ₂	N ₂	O ₂	F ₂	Cl ₂	Br ₂	I ₂											
1b	35	Chlorine 35 is the common isotope in sample as the average is closer to 35 than 37																
1c	One from: <table><tr><td>Fluorine</td><td>Bromine</td></tr><tr><td>Iodine</td><td>Astatine</td></tr></table>	Fluorine	Bromine	Iodine	Astatine	Elements in the same chemical group have similar chemical properties due to having the same number of outer electrons.												
Fluorine	Bromine																	
Iodine	Astatine																	
1d	<table><tr><td>10</td><td></td></tr><tr><td></td><td>20</td></tr></table>	10			20	<table><tr><td>²⁴₁₂Mg²⁺</td><td>No. of protons = atomic number = 12 No of neutrons = mass number - atomic number = 24 - 12 = 12 No of electrons = atomic number - charge = 12 - (2) = 10</td></tr><tr><td>³⁷₁₇Cl⁻</td><td>No. of protons = atomic number = 17 No of neutrons = mass number - atomic number = 37 - 17 = 20 No of electrons = atomic number - charge = 17 - (-1) = 18</td></tr></table>	²⁴ ₁₂ Mg ²⁺	No. of protons = atomic number = 12 No of neutrons = mass number - atomic number = 24 - 12 = 12 No of electrons = atomic number - charge = 12 - (2) = 10	³⁷ ₁₇ Cl ⁻	No. of protons = atomic number = 17 No of neutrons = mass number - atomic number = 37 - 17 = 20 No of electrons = atomic number - charge = 17 - (-1) = 18								
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2a	Hydrocarbon	A hydrocarbons is a compounds containing carbon and hydrogen only.																
2b(i)	Hydrogen	C ₂ H ₆ → C ₂ H ₄ + H ₂																
2b(ii)	decolourises bromine solution	Unsaturated compounds will decolourise bromine solution quickly. Bromine adds across the double bond in ethene to form 1,2-dibromoethane																
2c(i)	H-C≡C-H	Carbon has valency of 4 and makes four bonds per carbon atom Hydrogen has valency of 1 and make only one bond per hydrogen atom																
2c(ii)A	carbon dioxide and water	Ethyne is a hydrocarbon containing only carbon and hydrogen atoms. - Complete combustion of carbon produced carbon dioxide. - Complete combustion of hydrogen produced water.																
2c(ii)B	Gives out heat	Exothermic reactions are reactions where heat is given out to the surroundings. Endothermic reactions are reactions where heat is taken in from the surroundings																
3a	Group 2	Only Groups 2 contains metals only. Group 1 starts with the non-metal hydrogen. The remaining groups 3-0 all start with a non-metal at the top of the group																
3b(i)	KMnO ₄	<table><tr><td>Write down Symbols and valency below</td><td>Cross-Over arrows to work out formula</td><td>Work out chemical formula (Cancel down if necessary)</td></tr><tr><td> K MnO₄⁻ 1 1 </td><td> K MnO₄⁻ 1 1 </td><td>KMnO₄</td></tr></table>	Write down Symbols and valency below	Cross-Over arrows to work out formula	Work out chemical formula (Cancel down if necessary)	K MnO₄⁻ 1 1	K MnO₄⁻ 1 1	KMnO ₄										
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3b(ii)	Aluminium or zinc	<table><tr><td>Observation</td><td>Blinding white light</td><td>Bright light</td><td>Red glow with few sparks</td><td>Dull red glow</td></tr><tr><td>Metal</td><td>Magnesium</td><td>X</td><td>Iron</td><td>Copper</td></tr><tr><td>Reactivity</td><td colspan="4">High ▶ Low</td></tr></table> Conclusion: X must have reactivity between Magnesium and Iron ∴ 2 possible metals: aluminium or zinc	Observation	Blinding white light	Bright light	Red glow with few sparks	Dull red glow	Metal	Magnesium	X	Iron	Copper	Reactivity	High ▶ Low				
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Metal	Magnesium	X	Iron	Copper														
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3c	Hydrogen Burns with a pop	<table><tr><td>Gas</td><td>Hydrogen</td><td>Oxygen</td><td>Carbon Dioxide</td></tr><tr><td>Test</td><td>Burns with a pop</td><td>Relights a glowing splint</td><td>Turns limewater cloudy</td></tr></table>	Gas	Hydrogen	Oxygen	Carbon Dioxide	Test	Burns with a pop	Relights a glowing splint	Turns limewater cloudy								
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3d(i)	Voltage between 0.5 and 2.7	As Iron is between magnesium and tin on the electrochemical series, the voltage of an iron/copper cell will be between the voltage of a magnesium/copper cell and a tin/copper cell																
3d(ii)	Electrically conducting solutions containing ions	An electrolyte is an ion containing solution which completes the circuit in a cell and allows electricity to flow.																

3d(iii)	one answer from:	temperature	concentration	Type of electrode	Distance between electrodes	Surface area of electrodes										
4a	From air	Problem Solving: Extracting information from a passage														
4b	Lowers temperature or don't get used up	A catalyst speeds up a chemical reaction but does not get used up in the reaction. This often means that a chemical reaction can proceed at a lower temperature.														
4c	534.1	1 litre jet fuel made from 4700g CO ₂ 5 litres jet fuel made from 5x4700g CO ₂ = 23500g CO gfm CO ₂ = (1x12)+(2x16) = 12+32 = 44g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{23500}{44} = 534.1\text{mol}$														
5	Open Ended Question:	3 mark answer		2 mark answer		1 mark answer										
		Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.										
		<table><tr><th colspan="4">Candidate Marks</th></tr><tr><th>0 marks</th><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr><tr><td>10%</td><td>55%</td><td>26%</td><td>9%</td></tr></table>		Candidate Marks				0 marks	1 mark	2 marks	3 marks	10%	55%	26%	9%	
Candidate Marks																
0 marks	1 mark	2 marks	3 marks													
10%	55%	26%	9%													
6a(i)	carboxyl group	<table><tr><td>—O—H</td><td>$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>					—O—H	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$	hydroxyl group	carboxyl group						
—O—H	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$															
hydroxyl group	carboxyl group															
6a(ii)	Addition Polymerisation	C=C double bonds in monomer open up and join together to form polymer														
6a(iii)	Polymer diagram:	<table><tr><td>$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$</td><td>$\begin{array}{c} \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \\ \quad \quad \quad \quad \quad \\ \text{—C—C—C—C—C—C—} \\ \quad \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$</td><td>$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{—C—C—} \\ \quad \\ \text{H} \quad \text{H} \end{array}$</td></tr><tr><td>Monomer</td><td>Polymer</td><td>Repeating Unit</td></tr></table>					$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \\ \quad \quad \quad \quad \quad \\ \text{—C—C—C—C—C—C—} \\ \quad \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{—C—C—} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	Monomer	Polymer	Repeating Unit				
$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \\ \quad \quad \quad \quad \quad \\ \text{—C—C—C—C—C—C—} \\ \quad \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{—C—C—} \\ \quad \\ \text{H} \quad \text{H} \end{array}$														
Monomer	Polymer	Repeating Unit														
6b(i)	CsCl and Material A	Problem Solving: Prediction of value and completion of bar graph														
6b(ii)	Bar bigger than 18 and lower than 32	As strontium is the metal between calcium and barium in group 2, the prediction for SrCl ₂ would be between CaCl ₂ (18) and BaCl ₂ (32).														
6c(i)	measuring cylinder or pipette	Beakers are less accurate than measuring cylinders at measuring volumes of liquid. Pipettes are the most accurate device for measuring exact volumes.														
6c(ii)	Line Graph Showing:	1 mark		1 mark		1 mark	1 mark									
		One mark is awarded for a graph which shows points plotted rather than bars.		The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales		The axes of the graph have suitable labels and units.	All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.									
7a	Trisilane	Prefix	Mono-	Di-	Tri-	Tetra-	Penta-	Hexa-	Hepta-	Octa-						
		Meaning	1	2	3	4	5	6	7	8						
7b	Si ₅ H ₁₂	General Formula of silanes is Si _n H _{2n+2} in line with alkanes C _n H _{2n+2} For pentasilane, n=5 ∴ 2n+2 = (2x5)+2 = 12 ∴ Pentasilane formula = Si ₅ H ₁₂														

7c	185	Compound	Monosilane	Disilane	Trisilane	Tetrasilane	Pentasilane	Hexasilane
		Formula	SiH ₄	Si ₂ H ₆	Si ₃ H ₈	Si ₄ H ₁₀	Si ₅ H ₁₂	Si ₆ H ₁₄
		Boiling Point (°C)	-112	-15	53	108	153	-
		Difference		97	68	55	45	(35)
		Prediction (°C)	-	-	-	-	-	185
7d	One diagram from:							
7e	Stronger intermolecular forces due to longer silicon chain	1st Mark: pentasilane has stonger/larger forces of attraction. 2nd Mark: forces of attractions are intermolecular/between molecules The longer the silicon chain, the greater the number of atoms within the molecule. Bigger molecules have stronger intermolecular bonds between molecules which raises the boiling point of pentasilane over tetrasilane						
7f	3.1	gfm SiO₂ = (1×28)+(2×16) = 28+32 = 60g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{6}{60} = 0.1\text{mol}$ $7\text{Mg} + 2\text{SiO}_2 + 14\text{HCl} \longrightarrow \text{Si}_2\text{H}_6 + 7\text{MgCl}_2 + 4\text{H}_2\text{O}$ 2mol 1mol 0.1mol 0.05mol gfm Si₂H₆ = (2×28)+(6×1) = 56+6 = 62g mass = no. of mol × gfm = 0.05mol × 62 = 3.1g						
8a	Fluorapatite	Problem Solving: extracting information from a passage						
8b	C ₆ H ₁₂ O	As there are no metals in the formula, any order of the non-metal elements is acceptable although most tend to list C then H then O.						
8c(i)	Nitrogen	ADP has the formula NH ₄ H ₂ PO ₄ contains nitrogen and phosphorus. Both are essential elements for plant growth. The 3rd essential element for plant growth is potassium.						
8c(ii)	Soluble	Fertilisers must contain at least one of the three essential elements nitrogen, phosphorus or potassium and be soluble in water.						
8d	31.6	gfm H₃PO₄ = (3×1)+(1×31)+(4×16) = 3+31+64 = 98g $\% \text{ P} = \frac{\text{mass of P}}{\text{gfm}} \times 100 = \frac{31}{98} \times 100 = 31.6\%$						
8e	Filtration	Solid calcium sulfate can be separated from a liquid by filtration.						
8f	$\frac{1}{2}$ mol or 0.5mol	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ 1mol : 2mol $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ 1mol : $\frac{1}{2}$mol						
9a(i)	0.627	heat energy = specific heat capacity × mass × change in Temperature $E_h = C \times m \times \Delta T$ $E_h = 4.18 \times 0.01 \times 15$ $E_h = 0.627 \text{ kJ}$						
9a(ii)	One from:	No draft shield	Energy lost to surroundings			Temperature not consistent in water		
		Glass absorbs heat	Incomplete combustion					

9b	150	1g biscuit releases 20.9 kJ 30g biscuit releases $20.9 \text{ kJ} \times \frac{30}{1}$ = 627 kJ 4.18kJ equals 1 kilocalorie 627 kJ equals $1 \text{ kilocalorie} \times \frac{627}{4.18}$ = 150 kilocalorie																																						
10a(i)	electrolysis	As caesium is more reactive than potassium then electrolysis would be the method to extract caesium from its ore. <table><tr><td>Method</td><td colspan="2">Electrolysis</td><td colspan="2">Heat With Carbon</td><td colspan="2">Heat Alone</td></tr><tr><td rowspan="3">Metals Made This Way</td><td>Potassium</td><td>Sodium</td><td>Zinc</td><td>Iron</td><td>Mercury</td><td>Silver</td></tr><tr><td>Lithium</td><td>Calcium</td><td>Nickel</td><td>Tin</td><td>Gold</td><td>Platinum</td></tr><tr><td>Magnesium</td><td>Aluminium</td><td>Lead</td><td>Copper</td><td></td><td></td></tr><tr><td>Reason</td><td colspan="2">most reactive metals</td><td colspan="2">medium reactive metals</td><td colspan="2">least reactive metals</td></tr></table>						Method	Electrolysis		Heat With Carbon		Heat Alone		Metals Made This Way	Potassium	Sodium	Zinc	Iron	Mercury	Silver	Lithium	Calcium	Nickel	Tin	Gold	Platinum	Magnesium	Aluminium	Lead	Copper			Reason	most reactive metals		medium reactive metals		least reactive metals	
Method	Electrolysis		Heat With Carbon		Heat Alone																																			
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10a(ii)	Reduction	Metal ores contain metal ions which are reduced to produce metal atoms $\text{Cs}^+ + \text{e}^- \rightarrow \text{Cs}$																																						
10b(i)	${}^0_{-1}\text{e}$	Particle	Proton	Neutron	Electron	Alpha	Beta																																	
		Symbol	${}^1_1\text{p}$	${}^1_0\text{n}$	${}^0_{-1}\text{e}$	${}^4_2\text{He}$	${}^0_{-1}\text{e}$																																	
10b(ii)	Alpha particles cannot pass through paper	Radiation	Alpha		Beta		Gamma																																	
		Mass	4		0		No mass																																	
		Charge	2		-1		No charge																																	
		Stopped by	Paper		Aluminium		Thick lead																																	
		Deflection	Towards negative		Towards positive		No defection																																	
		Use	Smoke detectors		Measuring thickness of paper in paper mill		Radiotherapy cancer treatment																																	
10b(iii)A	One answer from:	Time taken for half the atoms in a sample to radioactively decay			Time for the radioactivity in. a sample to half																																			
10b(iii)B	$\frac{15}{16}$	Time	No. of half-lives	Fraction Remaining	% Remaining = $\frac{1}{16}$ % Decayed = $1 - \frac{1}{16}$ = $\frac{15}{16}$ % Remaining = 6.25% % Decayed = $100\% - 6.25\%$ = 93.75%																																			
		0 years	0	1 (100%)																																				
		30 years	1	$\frac{1}{2}$ (50%)																																				
		60 years	2	$\frac{1}{4}$ (25%)																																				
		90 years	3	$\frac{1}{8}$ (12.5%)																																				
		120 years	4	$\frac{1}{16}$ (6.25%)																																				
11a	covalent molecular	Tungsten is a metal. Metals and non-metal compounds usually form ionic compounds. All ionic compounds are solid at room temperature. Tungsten(VI) fluoride is a gas at room temperature and this means that tungsten(VI) fluoride must be covalent molecular.																																						
11b(i)	$\text{WF}_6 + 3\text{H}_2\text{O} \rightarrow \text{WO}_3 + 6\text{HF}$	$\text{WF}_6 + 3\text{H}_2\text{O} \rightarrow \text{WO}_3 + 6\text{HF}$																																						
11b(ii)	concentration of hydrogen ions greater than concentration of hydroxide ions	Type	Relationship between Concentration of Hydrogen ions and Hydroxide Ions																																					
		acid	Concentration of hydrogen ions greater than		Concentration of hydroxide ions																																			
		neutral	Concentration of hydrogen ions equal to		Concentration of hydroxide ions																																			
		alkali	Concentration of hydrogen ions less than		Concentration of hydroxide ions																																			
11c	$\text{W}^{6+} + 2\text{e}^- \rightarrow \text{W}^{4+}$	When balancing an ion-electron equation, adding electrons will balance the charge by adding to the most positive side of the equation.																																						
12	Open Ended Question: <table><tr><td colspan="4">Candidate Marks</td></tr><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>15%</td><td>59%</td><td>17%</td><td>9%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	15%	59%	17%	9%	3 mark answer Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		2 mark answer Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		1 mark answer Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																						
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2023 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	26	9	2	62	1	0
2	A	92	2	1	5	1	0
3	C	3	9	80	7	1	0
4	B	14	58	17	11	1	0
5	B	5	52	13	29	1	0
6	D	13	5	11	71	1	0
7	B	15	54	15	14	1	0
8	C	19	12	45	24	1	0
9	B	5	59	11	23	1	0
10	A	49	27	14	9	1	0
11	C	30	7	56	7	1	0
12	B	30	33	16	20	1	0
13	A	87	5	5	4	1	0
14	D	12	13	7	68	1	0
15	D	6	5	11	78	1	0
16	C	19	8	69	3	1	0
17	A	72	2	19	6	1	0
18	B	2	70	26	2	1	0
19	D	5	4	4	86	1	0
20	A	29	55	8	7	1	0
21	C	11	8	72	9	1	0
22	C	9	12	74	5	1	0
23	C	6	13	70	11	1	0
24	A	80	13	4	2	1	0
25	B	14	44	28	13	1	0

2023 Nat5 Chemistry Section 2 Marks Data Analysis

2023 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	1	0.7	71.6%	39	1.6%
Q1b	1	0.8	78.0%	44	1.6%
Q1c	1	0.8	81.3%	44	2.0%
Q1d	2	1.2	60.2%	56	0.5%
Q2a	1	0.8	83.0%	46	2.7%
Q2bi	1	0.7	68.8%	52	3.8%
Q2bii	1	0.5	48.4%	52	9.4%
Q2ci	1	0.7	67.1%	52	3.1%
Q2ciiA	1	0.5	48.7%	56	4.4%
Q2ciiB	1	0.9	91.4%	36	1.1%
Q3a	1	0.9	89.9%	33	0.9%
Q3bi	1	0.5	53.7%	49	4.0%
Q3bii	1	0.6	63.0%	36	2.8%
Q3c	2	1.1	53.0%	57	12.1%
Q3di	1	0.9	85.6%	37	4.8%
Q3dii	1	0.3	25.0%	42	8.8%
Q3diii	1	0.6	61.8%	34	2.8%
Q4a	1	1	96.2%	24	0.9%
Q4b	1	0.7	71.6%	38	1.6%
Q4c	2	1	48.9%	58	4.6%
Q5	3	1.3	44.7%	66	6.6%
Q6ai	1	0.7	65.1%	56	5.5%
Q6aai	1	0.6	62.9%	34	3.7%
Q6aiii	1	0.5	47.4%	49	5.9%
Q6bi	1	0.4	41.9%	43	1.1%
Q6bii	1	0.8	76.5%	32	5.3%
Q6ci	1	0.9	85.0%	33	3.9%
Q6cii	4	2.9	72.5%	44	0.8%
Q7a	1	0.8	82.4%	35	2.1%
Q7b	1	0.9	93.9%	25	1.4%
Q7c	1	0.5	48.3%	22	1.7%
Q7d	1	0.7	68.0%	54	4.5%
Q7e	2	1	51.9%	51	2.6%
Q7f	3	1.5	51.4%	68	10.1%
Q8a	1	1	96.1%	29	1.1%
Q8b	1	0.4	37.7%	53	5.5%
Q8ci	1	0.6	62.1%	48	1.7%
Q8cii	1	0.8	79.3%	54	6.9%
Q8d	3	2.4	81.2%	57	5.1%
Q8e	1	0.3	31.0%	36	9.4%
Q8f	1	0.5	51.3%	54	10.2%
Q9ai	3	2	65.8%	67	6.0%
Q9aai	1	0.5	52.2%	57	9.1%
Q9b	2	1.3	67.3%	54	6.5%
Q10ai	1	0.6	58.7%	53	7.6%
Q10aai	1	0.4	44.7%	54	11.1%
Q10bi	1	0.5	48.1%	48	8.8%
Q10bii	1	0.8	76.8%	51	5.7%
Q10biiiA	1	0.3	31.3%	40	4.3%
Q10biiiB	3	1.6	53.2%	67	7.0%
Q11a	1	0.2	24.6%	7	3.7%
Q11bi	1	0.8	79.0%	52	3.7%
Q11bii	1	0.6	62.0%	63	12.1%
Q11c	1	0.4	43.8%	54	14.5%
Q12	3	1.2	40.3%	64	7.6%
Paper	75	46	61.3%		

2023 Nat5 Chemistry Section 2 Distribution of Marks Analysis

2023 Item	0	1	2	3	4
Q1a	28	72			
Q1b	22	78			
Q1c	19	81			
Q1d	27	26	47		
Q2a	17	83			
Q2bi	31	69			
Q2bii	52	48			
Q2ci	33	67			
Q2ciiA	51	49			
Q2ciiB	9	91			
Q3a	10	90			
Q3bi	46	54			
Q3bii	37	63			
Q3c	33	28	39		
Q3di	14	86			
Q3dii	75	25			
Q3diii	38	62			
Q4a	4	96			
Q4b	28	72			
Q4c	38	26	36		
Q5	10	55	26	9	
Q6ai	35	65			
Q6aii	37	63			
Q6aiii	53	47			
Q6bi	58	42			
Q6bii	23	77			
Q6ci	15	85			
Q6cii	2	5	20	50	24
Q7a	18	82			
Q7b	6	94			
Q7c	52	48			
Q7d	32	68			
Q7e	33	31	36		
Q7f	34	11	21	34	
Q8a	4	96			
Q8b	62	38			
Q8ci	38	62			
Q8cii	21	79			
Q8d	11	7	10	72	
Q8e	69	31			
Q8f	49	51			
Q9ai	19	4	36	40	
Q9aii	48	52			
Q9b	20	25	55		
Q10ai	41	59			
Q10aii	55	45			
Q10bi	52	48			
Q10bii	23	77			
Q10biiiA	69	31			
Q10biiiB	23	16	40	21	
Q11a	75	25			
Q11bi	21	79			
Q11bii	38	62			
Q11c	56	44			
Q12	15	59	17	9	